

Lab 1 — Algorithms under attack

TC1037 Algorithmic Thinking · Group 510 · Friday 25 September 2026,
16:30–17:30, Room 4206

Why this lab is paper-only

Our lab runs one hour **before** the Friday session that teaches Python, so nobody is asked to write Python today. This lab gives Tuesday's idea a second, harder pass and hands you into the Friday lecture with a concrete failure fresh in mind. Coding starts in Lab 2 (Friday 2 October).

Running order

	Minutes	What
1	20	Initial prior-knowledge diagnostic — administered by the professor at 16:30. No built-in timer: the professor calls time at 20 minutes. You need your laptop for this part.
2	3	Settle, hand out paper, collect Tuesday's algorithms.
3	32	The two exercises below.

Bring **laptop, notebook and pencil** — laptop for the diagnostic, paper for everything after it. Also bring your algorithm from Tuesday's Exercise 1 (price validation); if you do not have it, you rewrite it, which takes four minutes and is not a penalty.

Exercise 1 — Break someone else's algorithm

Pairs. 15 minutes.

1. Swap your Tuesday price-validation algorithm with your partner.
2. **Execute theirs literally**, on paper, for each of these four inputs — writing down what their steps actually produce, not what you know is meant:

	Input
a	50
b	0
c	-3
d	holà

3. For every input where their algorithm does something wrong, undefined, or unstated, write **one line** naming the step number and the problem.

Hand back. You now have a bug report on your own algorithm, written by someone who was not trying to be kind to it.

The point: nobody's Tuesday version survives all four. Input 0 is the interesting one — “positive” either includes it or does not, and almost nobody said which.

Exercise 2 — The algorithm that must terminate

Individual. 17 minutes.

A vending machine holds a drink costing **12 pesos**. It accepts coins of 1, 2, 5 and 10 pesos, one at a time. It must keep accepting coins until the customer has paid enough, then release the drink and return any change.

Write the algorithm. INPUT: / OUTPUT: lines, numbered steps.

Then answer, in writing, underneath:

- **Where exactly does your algorithm stop?** Point at the step number.
- What does it do if the customer pays 12 exactly?
- What does it do if the customer puts in a 10 and then walks away?

You do not have the tools to write this as a program yet — that is fine. This is the algorithm you will be able to write as a program in Week 3, and I want it on paper first.

Submission

Hand in, before you leave (to Luis, the TA):

- Your two handwritten algorithms.
- Your partner's bug report on your Exercise 1 (with your name on it).
- Exercise 2's three written answers.

Nothing is submitted to Canvas from this lab. Paper only, in the room.
